

Jeffrey Lazar

MASSACHUSETTS INSTITUTE OF TECHNOLOGY · POSTDOCTORAL ASSOCIATE

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Summary

I am a postdoctoral associate in the Laboratory for Nuclear Science at the Massachusetts Institute of Technology. My work broadly centers around high-energy neutrino astrophysics, especially the IceCube, KM3NeT, and TAMBO observatories. My primary work is in Beyond-Standard-Model phenomenology, Monte Carlo simulation, and novel searches for new physics; however, my interests are broad, and I am always open to trying new ideas.

Education

University of Wisconsin–Madison

Ph.D in Physics

Madison, WI

Sep. 2017–Oct. 2023

Thesis: *Searching for solar neutrinos and building an open-source future for neutrino astronomy*. Advised by Francis Halzen and Carlos A.

Argüelles-Delgado.

Columbia University

B.S. in Physics

New York, NY

Sep. 2013–May 2017

B.S. in Mathematics

Sep. 2013–May 2017

Professional Experience

Massachusetts Institute of Technology

Postdoctoral Associate

Cambridge, MA

2026–Present

Postdoctoral associate in the Laboratory for Nuclear Science.

Université Catholique de Louvain

Postdoctoral Fellow

Louvain-la-Neuve, Belgium

Nov. 2023–2026

Postdoctoral fellow funded by the Belgian American Educational Foundation and Le Fonds de la Recherche Scientifique working in Professor de Wasseige's group. My focuses are searches for new neutrino signatures in the IceCube and KM3NeT neutrino telescopes and simulation of high-energy neutrinos.

University of Wisconsin–Madison

Research Assistant

Madison, WI

Sep. 2017–Oct. 2023

Research assistant on IceCube experiment under the supervision of Francis Halzen and co-supervision of Carlos Argüelles with a focus on dark matter and beyond Standard Model neutrino physics, neutrino simulation, and machine learning.

The NSF AI Institute for Artificial Intelligence and Fundamental Interactions (IAIFI)

Student Research Associate

Cambridge, MA

Jan. 2022–Oct. 2023

Helped plan IAIFI machine learning summer school.

Columbia University

Research Assistant

New York, NY

Sep. 2015–May 2017

Research assistant on the LIGO experiment under the supervision of Szabolcs Marka with a focus on timing system calibration

Teaching Experience

Université Catholique de Louvain

Guest Lecturer

Louvain-la-Neuve, Belgium

May 2022

Gave lectures on indirect detection of dark matter using astroparticle telescopes to masters-level students.

University of Wisconsin–Madison

Teaching Assistant

Madison, WI

Sep. 2017–May 2018

Teaching assistant for introductory mechanics and electromagnetism courses

Naval Reserve Officers' Training Corps

Physics tutor

Madison, WI

Sep. 2018–May 2020

Tutor for all members of the NROTC program covering topics from introductory mechanics to general relativity

Independent Physics Tutor

Madison, WI

Physics and math tutor

Sep. 2017–May 2020

Tutor for undergraduate and high school students. I taught physics ranging from introductory mechanics to advanced quantum and classical mechanics. I additionally taught calculus and linear algebra.

Mentoring Experience

Université Catholique de Louvain Mentor Mentoring Perrine Wilmet, a master's student, on an effort to combine observations across multiple detectors to improve sensitivity to astrophysical neutrino sources.	<i>Louvain-la-Neuve, Belgium</i> <i>May 2024–Present</i>
Harvard University Student Mentor Mentored David Kim, a second-year student who visited from Cornell over the summer and whose research continued into the school year. This work resulted in a contribution to the International Cosmic Ray Conference, including a proceeding on which David was the lead author.	<i>Cambridge, MA</i> <i>Jun. 2022–Sep. 2023</i>
Harvard University Student Mentor Mentored Jorge Garcia-Ponce, a first-year undergraduate associated with the Argüelles-Delgado group. We are working to develop the official simulation for the proposed Tau Air-Shower Mountain-Based Observatory.	<i>Cambridge, MA</i> <i>May 2021–Dec. 2022</i>
Harvard University Student Mentor Mentoring Santiago Giner, a sophomore undergraduate associated with the Argüelles-Delgado group. We are working on a quantum computing-based machine learning classifier to differentiate different event morphologies in the IceCube detector. To this end, we have implemented a quantum random access protocol to store classical information in a quantum system and developed an open-source simulation of the IceCube detector. This work led to presentations at the American Physics Society's April meeting and at the 30 th International Conference on Neutrino Physics and Astrophysics.	<i>Cambridge, MA</i> <i>May 2021–Oct. 2023</i>
Harvard University Student Mentor Mentored Josh Villarreal, a senior undergraduate associated with the Argüelles-Delgado group. We are looking to develop a new event selection, designed to optimize sensitivity to solar atmospheric neutrinos. This has lead to Josh giving talks on the topic of solar atmospheric neutrinos at the XIX th International Workshop on Neutrino Telescopes, the 9 th Very Large Volume Neutrino Telescope Workshop, and the 37 th International Cosmic Ray Conference.	<i>Cambridge, MA</i> <i>Jan. 2021–Sep. 2021</i>
Harvard University Student Mentor Mentored Pablo Castaño, a first-year undergraduate associated with the Argüelles-Delgado group. We are looking to develop error-tolerant, neutrino oscillation algorithms on a quantum computer. This has included teaching basic quantum computing as well as basic quantum mechanics. This mentoring has lead to Pablo giving a talk at the APS meeting.	<i>Cambridge, MA</i> <i>Nov. 2020–Jun. 2021</i>

Awards and Fellowships

Le Fonds de la Recherche Scientifique Université Catholique de Louvain Post-doctoral research fellow at the Université catholique de Louvain.	<i>Louvain-la-Neuve, Belgium</i> <i>Sep. 2024–Present</i>
Belgian American Educational Foundation Université Catholique de Louvain Post-doctoral research fellow at the Université Catholique de Louvain.	<i>Louvain-la-Neuve, Belgium</i> <i>Nov. 2023–Aug. 2024</i>
Fellow of the Physics Department Harvard University Visiting fellow of the physics department working with Carlos Argüelles on the IceCube experiment.	<i>Cambridge, MA</i> <i>May 2020–Oct. 2023</i>
New Graduate Student Award University of Wisconsin–Madison Received an award presented to top incoming graduate students by the physics department.	<i>Madison, WI</i> <i>April 2017</i>

Community Service and Leadership Roles

IceCube Early Career Scientists Early Career Scientist Representative Representative to the IceCube Board on behalf of IceCube graduate students and postdocs.	<i>Jan. 2026–Present</i>
IAIFI Summer School Local Committee Member Planned the first summer school hosted by the Institute for Artificial Intelligence and Fundamental Interactions.	<i>Cambridge, MA</i> <i>Oct. 2021–Aug. 2022</i>

IceCube Beyond Standard Model Working Group

Technical Lead

May 2022–Present

Coordinate technical aspects related to Beyond Standard Model working group.

IceCube Beyond Standard Model Working Group

Data Curator

Apr. 2021–May 2022

Coordinate standards and checks to ensure all analyses are reproducible.

Outreach and Community Involvement

IceCube Master Class

Cambridge, MA

Planning Committee Member and lecturer

Jan. 2023–May 2023

Planned a day-long class that taught high-school students basic particle physics and programming in the context of the IceCube Observatory. I developed the analysis activity, which used open-source tools to find the Moon's shadow in IceCube.

WIPAC MasterClass

Madison, WI

Instructor

Apr. 2021

Lead students through a cosmic-ray analysis in which they calculated the energy-dependence of the cosmic-ray flux using IceTop data.

WIPAC High School Internship

Madison, WI

Student Mentor

Oct. 2019–Jan. 2020

Worked with high school students as part of an after school science program. This included leading exercises, presenting research at an appropriate level, and answering questions about my research and navigating an undergraduate physics program

Dark Matter Day

Madison, WI

Presenter

Oct. 2020

Presented my research to the general public as part of IceCube's participation in the larger Dark Matter Day initiative.

Columbia University Project for the Homeless

New York, NY

Head Coordinator

May 2015–May 2017

Coordinated day-to-day activities for the homeless shelter at Stephen Wise Free Synagogue (SWFS). Coordinated with overnight volunteers for the shelters at SWFS and the New York Society for Ethical Culture. Stayed overnight in the event that a volunteer was not available

Invited Seminars and Plenary Talks

Gent Particle Physics Seminar

Gent, Belgium

Presented a seminar on the proposed, next-generation Tau Air-Shower Mountain-Based Observatory

Mar. 2026

Corfu Workshop on the Standard Model and Beyond

Corfu, Greece

Recent Results from the IceCube Neutrino Observatory

Aug. 2025

OSU Center for Cosmology and Astroparticle Physics

Columbus, OH

Presented a seminar on the proposed, next-generation Tau Air-Shower Mountain-Based Observatory

Apr. 2025

Wisconsin IceCube Astroparticle Physics Center

Madison, WI

Presented a seminar on the proposed, next-generation Tau Air-Shower Mountain-Based Observatory

Mar. 2025

University of Science and Technology of China

Hefei, China

Presented a seminar on the proposed, next-generation Tau Air-Shower Mountain-Based Observatory

Nov. 2024

8th Shanghai Particle Physics and Cosmology Symposium

Shanghai, China

Presented an invited plenary talk on open-source tools and methods for Neutrino telescopes

Nov. 2024

Tsung-Dao Lee Institute High-Energy Physics Seminar

Shanghai, China

Presented a seminar on the proposed, next-generation Tau Air-Shower Mountain-Based Observatory

Nov. 2024

CERN Quantum Computing Seminar

Geneva, Switzerland

Presented a seminar on quantum random access embedding.

Oct. 2024

Invisibles Workshop 2024

Bologna, Italy

Presented invited plenary talk on recent results from the IceCube and KM3NeT Neutrino Observatories

Jul. 2024

Cosmic-Ray International Studies and Multi-messenger Astroparticle Conference

Trapani, Italy

Presented invited plenary talk on recent results from the IceCube Neutrino Observatory

Jun. 2024

Durham University

Durham, UK

Presented a seminar on the proposed, next-generation Tau Air-Shower Mountain-Based Observatory

May 2024

Niels Bohr Institute

Copenhagen, Denmark

Presented a seminar on the proposed, next-generation Tau Air-Shower Mountain-Based Observatory

Apr. 2024

Instituto de Física Corpuscular (IFIC)

Presented a seminar on the proposed, next-generation Tau Air-Shower Mountain-Based Observatory

Valencia, Spain

Mar. 2024

European Organization for Nuclear Research (CERN)

Presented a seminar on highlights of recent IceCube dark matter results.

Geneva, Switzerland

Feb. 2024

Harvard University Laboratory for Particle Physics and Cosmology

Presented a seminar with results of the TAMBO Monte Carlo simulation.

Cambridge, MA

Feb. 2024

Institute for Artificial Intelligence and Fundamental Interactions

Presented a seminar on open-source simulation and machine learning in neutrino telescopes.

Cambridge, MA

Sep. 2023

American Physical Society April Meeting

Presented invited parallel talk on beyond Standard Model physics with neutrinos above 100 GeV

Minneapolis, MN

Apr. 2023

International Conference on the Physics of the Two Infinities

Presented invited plenary talk on neutrino sources and beyond Standard Model searches with IceCube Neutrino Observatory

Kyoto, Japan

Mar. 2023

Instituto de Física Corpuscular

Presented seminar on beyond Standard Model searches in current and future neutrino observatories.

Valencia, Spain

Mar. 2022

The Laboratory for Nuclear Science

Presented seminar on high-energy solar neutrinos to local research group

Cambridge, MA

Mar. 2022

The Center for Cosmology and Astroparticle Physics

Presented seminar on high-energy solar neutrinos to local research group

Columbus, OH

Feb. 2022

Kavli Institute for the Physics and Mathematics of the Universe

Presented seminar on high-energy solar neutrinos to local research group

Tokyo, Japan

Nov. 2021

The XXVIII International Conference on Supersymmetry and the Unification of Fundamental Interactions

Presented an invited parallel talk on the recent results of IceCube's high-energy solar WIMP search

Shanghai, China

Aug. 2021

European Physical Society conference on high-energy Physics 2021

Presented an invited parallel talk on the current status of IceCube's dark matter searches

Hamburg, Germany

Jul. 2021

37th International Cosmic Ray Conference

Presented a highlight plenary talk on the current status of IceCube's solar WIMP search

Berlin, Germany

Jul. 2021

Fermilab Cosmic Physics Center

Presented seminar on high-energy solar neutrinos to local research group

Batavia, IL

Jun. 2021

AstroParticle and Cosmology Laboratory

Presented seminar on high-energy solar neutrinos to local research group

Paris, France

May 2021

Selected Publications

My ORCID profile can be found [here](#). For a full list of publications, please visit my [Inspire profile](#). The following are publications, proceedings, and whitepapers to which I made significant and essential contributions.

Improving Neutrino Point Source Sensitivity with Source-Informed Event Selection

J. Lazar, C. A. Argüelles, and P. Zhelnin

Submitted to *Phys. Rev. Lett.*

arXiv:2604.06509

CORSIKA 8: A General Framework for Particle Cascade Simulations

J. M. Alameddine, *et al.* (CORSIKA 8)

Submitted to *Astropart. Phys.*

arXiv:2604.01850

NuBench: An Open Benchmark for Deep Learning-Based Event Reconstruction in Neutrino Telescopes

R. Ørsøe, S. Meighen-Berger, J. Lazar, J. Prado, I. Mozún-Mateo, A. Rosted, P. Weigel, A. Llorente Anaya

JINST

arXiv:2511.13111

Search for High-Energy Neutrinos From the Sun Using Ten Years of IceCube Data

R. Abbasi *et al.* (IceCube)

Submitted to *Phys. Rev. Lett.*

arXiv:2507.08457

Measuring the High-Energy Neutrino Sky using the Deep-Valley Neutrino Observatory TAMBO

C. A. Argüelles *et al.* (TAMBO)

Nature Astron.

arXiv:2507.08070

Supernovae Time Profiles as a Probe of New Physics at Neutrino Telescopes

J. Lazar, Y. Li, C. A. Argüelles, V. Brdar

Phys. Rev. Lett.

arXiv:2403.09781

New Pathways in Neutrino Physics via Quantum-Encoded Data Analysis

J. Lazar, S. Giner, G. Gatti, C. A. Argüelles, M. Sanz

Phys. Rev. Res.

arXiv:2402.19306

Prometheus: An Open-Source Neutrino Telescope Simulation

J. Lazar, S. Meighen-Berger, C. Haack, D. Kim, S. Giner, C. A. Argüelles

Comp. Phys. Com.

arXiv:2304.14526

Trigger-Level Event Reconstruction for Neutrino Telescopes Using Sparse Submanifold Convolutional Neural Networks

F. Yu, J. Lazar, C. A. Argüelles

Phys. Rev. D.

arXiv:2303.08812

TauRunner: A Public Python Program to Propagate Neutral and Charged Leptons

I. Sefa, J. Lazar, A. Pizzuto, O. Vasquez, C. A. Argüelles, J. Vandenbroucke

Comp. Phys. Com.

arXiv:2110.14662

χ_{arovi} : a tool for neutrino flux generation from WIMPs

Q. Liu, J. Lazar, C. A. Argüelles, A. Kheirandish

JCAP

arXiv:2007.15010

Current status of simulation for the Tau Air-Shower Mountain-Based Observatory

J. Lazar and P. Zhelnin

PoS ICRC 2023

Prometheus: An Open-Source Neutrino Telescope Simulation

D. Kim and J. Lazar

PoS ICRC 2023

arXiv:2307.16432

All-Energy Search for Solar Atmospheric Neutrinos with IceCube

J. Lazar and I. Martinez-Soler

PoS ICRC 2023

Recent Progress in Solar Atmospheric Neutrino Searches with IceCube

J. Villarreal, G. Roellinghoff, J. Lazar

PoS ICRC 2021

arXiv:2107.13696

Searching for Dark Matter from the Sun with Ten Years of IceCube

J. Lazar, Q. Liu, C. A. Argüelles, A. Kheirandish

PoS ICRC 2019

Search for Dark Matter Annihilation to Neutrinos from the Sun

C. A. Argüelles, A. Kheirandish, J. Lazar, Q. Liu

PoS ICRC 2019

arXiv:1909.03930

Quantum Computing for Data Analysis in High-Energy Physics

A. Delgado, *et al.*

Snowmass Whitepaper

arXiv:2203.08805

Observing the High-Energy Sun

J. Lazar and C. A. Argüelles

Snowmass Letters of Interest

SnowMass 21

Tau Neutrinos in the Next Decade: from GeV to EeV

R. M. Abraham *et al.*

Snowmass Whitepaper

Journal of Physics G